Perform the following operations using Python on the Air quality data sets

- Data cleaning
- Data integration
- Data transformation
- Error correcting
- Data model building

In [6]:
```
# import pandas library
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import LabelEncoder, MinMaxScaler
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score
```

In [7]:
```
# Reading csv file
df = pd.read_csv('Air Quality.csv', encoding='cp1252')
```

In [8]:
```
df.head()
```

Out[8]:

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC)
0	10/03/2004	18:00:00	2.6	1360.0	150.0	11.9	1046.0
1	10/03/2004	19:00:00	2.0	1292.0	112.0	9.4	955.0
2	10/03/2004	20:00:00	2.2	1402.0	88.0	9.0	939.0
3	10/03/2004	21:00:00	2.2	1376.0	80.0	9.2	948.0
4	10/03/2004	22:00:00	1.6	1272.0	51.0	6.5	836.0

In [12]:
```
# Information about each column data
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9471 entries, 0 to 9470
Data columns (total 17 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Date           9357 non-null   object
 1   Time           9357 non-null   object
 2   CO(GT)         9357 non-null   float64
 3   PT08.S1(CO)    9357 non-null   float64
 4   NMHC(GT)       9357 non-null   float64
 5   C6H6(GT)       9357 non-null   float64
 6   PT08.S2(NMHC)  9357 non-null   float64
 7   NOx(GT)        9357 non-null   float64
 8   PT08.S3(NOx)   9357 non-null   float64
 9   NO2(GT)        9357 non-null   float64
 10  PT08.S4(NO2)   9357 non-null   float64
 11  PT08.S5(O3)    9357 non-null   float64
 12  T              9357 non-null   float64
 13  RH             9357 non-null   float64
 14  AH             9357 non-null   float64
 15  Unnamed: 15    0 non-null      float64
 16  Unnamed: 16    0 non-null      float64
dtypes: float64(15), object(2)
memory usage: 1.2+ MB
```

In [11]:
```
df.columns
```

Out[11]:
```
Index(['Date', 'Time', 'CO(GT)', 'PT08.S1(CO)', 'NMHC(GT)', 'C6H6(GT)',
       'PT08.S2(NMHC)', 'NOx(GT)', 'PT08.S3(NOx)', 'NO2(GT)', 'PT08.S4(NO2)',
       'PT08.S5(O3)', 'T', 'RH', 'AH', 'Unnamed: 15', 'Unnamed: 16'],
      dtype='object')
```

---

a. Data Cleaning

In [7]:
```
# Drop empty trailing columns
df.drop(columns=['Unnamed: 15', 'Unnamed: 16'], inplace=True, errors='ignore')
```

In [8]:
```
# Replace -200 (sensor error sentinel used in UCI dataset) with NaN
df.replace(-200, np.nan, inplace=True)
df.replace(-200.0, np.nan, inplace=True)
```

In [9]:
```
# Optimise dtypes for memory
float32_cols = ['CO(GT)', 'C6H6(GT)', 'T', 'RH', 'AH']
for col in float32_cols:
    df[col] = df[col].astype('float32')

df['Date'] = df['Date'].astype('string')
df['Time'] = df['Time'].astype('string')

df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9471 entries, 0 to 9470
Data columns (total 15 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Date           9357 non-null   string
 1   Time           9357 non-null   string
 2   CO(GT)         7674 non-null   float32
 3   PT08.S1(CO)    8991 non-null   float64
 4   NMHC(GT)       914 non-null    float64
 5   C6H6(GT)       8991 non-null   float32
 6   PT08.S2(NMHC)  8991 non-null   float64
 7   NOx(GT)        7718 non-null   float64
 8   PT08.S3(NOx)   8991 non-null   float64
 9   NO2(GT)        7715 non-null   float64
 10  PT08.S4(NO2)   8991 non-null   float64
 11  PT08.S5(O3)    8991 non-null   float64
 12  T              8991 non-null   float32
 13  RH             8991 non-null   float32
 14  AH             8991 non-null   float32
dtypes: float32(5), float64(8), string(2)
memory usage: 925.0 KB
```

In [10]:
```
df.drop_duplicates(inplace=True)
print('Shape after dropping duplicates:', df.shape)
```

```
Shape after dropping duplicates: (9358, 15)
```

In [11]:
```
df.isna().sum()
```

Out[11]:
```
Date                1
Time                1
CO(GT)           1684
PT08.S1(CO)       367
NMHC(GT)         8444
C6H6(GT)          367
PT08.S2(NMHC)     367
NOx(GT)          1640
PT08.S3(NOx)      367
NO2(GT)          1643
PT08.S4(NO2)      367
PT08.S5(O3)       367
T                 367
RH                367
AH                367
dtype: int64
```

In [12]:
```
percent_missing = df.isnull().sum() * 100 / len(df)
percent_missing.sort_values(ascending=False)
```

Out[12]:
```
NMHC(GT)         90.232956
CO(GT)           17.995298
NO2(GT)          17.557170
NOx(GT)          17.525112
PT08.S2(NMHC)     3.921778
C6H6(GT)          3.921778
PT08.S1(CO)       3.921778
PT08.S5(O3)       3.921778
T                 3.921778
PT08.S3(NOx)      3.921778
PT08.S4(NO2)      3.921778
RH                3.921778
AH                3.921778
Date              0.010686
Time              0.010686
dtype: float64
```

In [13]:
```
# Fill categorical with mode, numeric with mean
for col in df.columns:
    if df[col].dtype == 'object' or str(df[col].dtype) == 'string':
        df[col] = df[col].fillna(df[col].mode()[0])
    else:
        df[col] = df[col].fillna(df[col].mean())
```

In [14]:
```
print('Missing values after cleaning:')
print(df.isna().sum())
```

```
Missing values after cleaning:
Date             0
Time             0
CO(GT)           0
PT08.S1(CO)      0
NMHC(GT)         0
C6H6(GT)         0
PT08.S2(NMHC)    0
NOx(GT)          0
PT08.S3(NOx)     0
NO2(GT)          0
PT08.S4(NO2)     0
PT08.S5(O3)      0
T                0
RH               0
AH               0
dtype: int64
```

In [15]:
```
df.head()
```

Out[15]:

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC)
0	10/03/2004	18:00:00	2.6	1360.0	150.0	11.9	1046.0
1	10/03/2004	19:00:00	2.0	1292.0	112.0	9.4	955.0
2	10/03/2004	20:00:00	2.2	1402.0	88.0	9.0	939.0
3	10/03/2004	21:00:00	2.2	1376.0	80.0	9.2	948.0
4	10/03/2004	22:00:00	1.6	1272.0	51.0	6.5	836.0

---

b. Data Integration

In [16]:

```
# Subset 1: Temporal identifiers
subSet1 = df[['Date', 'Time']]

# Subset 2: Environmental / meteorological readings
subSet2 = df[['T', 'RH', 'AH']]

subSet1.head()
```

Out[16]:

	Date	Time
0	10/03/2004	18:00:00
1	10/03/2004	19:00:00
2	10/03/2004	20:00:00
3	10/03/2004	21:00:00
4	10/03/2004	22:00:00

In [17]:

```
subSet2.head()
```

Out[17]:

	T	RH	AH
0	13.6	48.900002	0.7578
1	13.3	47.700001	0.7255
2	11.9	54.000000	0.7502
3	11.0	60.000000	0.7867
4	11.2	59.599998	0.7888

In [18]:

```
# Horizontally integrate the two subsets
concatenated_df = pd.concat([subSet1, subSet2], axis=1)
concatenated_df.head()
```

Out[18]:

	Date	Time	T	RH	AH
0	10/03/2004	18:00:00	13.6	48.900002	0.7578
1	10/03/2004	19:00:00	13.3	47.700001	0.7255
2	10/03/2004	20:00:00	11.9	54.000000	0.7502
3	10/03/2004	21:00:00	11.0	60.000000	0.7867
4	10/03/2004	22:00:00	11.2	59.599998	0.7888

In [19]:

```
# Merge pollutant subsets with environmental readings using index
pollutants = df[['CO(GT)', 'NOx(GT)', 'NO2(GT)', 'C6H6(GT)']]
env        = df[['T', 'RH', 'AH']]
merged_df  = pd.merge(pollutants, env, left_index=True, right_index=True)
merged_df.head()
```

Out[19]:

	CO(GT)	NOx(GT)	NO2(GT)	C6H6(GT)	T	RH	AH
0	2.6	166.0	113.0	11.9	13.6	48.900002	0.7578
1	2.0	103.0	92.0	9.4	13.3	47.700001	0.7255
2	2.2	131.0	114.0	9.0	11.9	54.000000	0.7502
3	2.2	172.0	122.0	9.2	11.0	60.000000	0.7867
4	1.6	131.0	116.0	6.5	11.2	59.599998	0.7888

---

c. Data Transformation

In [20]:

```
# Label-encode categorical columns
encoder = LabelEncoder()
cat_cols = ['Date', 'Time']
```

```python
for col in cat_cols:
    df[col] = encoder.fit_transform(df[col].astype(str))

df[cat_cols].head()
```

Out[20]:

	Date	Time
0	114	18
1	114	19
2	114	20
3	114	21
4	114	22

In [21]:

```python
# Min-Max Normalisation on numeric columns
num_cols = ['CO(GT)', 'PT08.S1(CO)', 'NMHC(GT)', 'C6H6(GT)',
            'PT08.S2(NMHC)', 'NOx(GT)', 'PT08.S3(NOx)', 'NO2(GT)',
            'PT08.S4(NO2)', 'PT08.S5(O3)', 'T', 'RH', 'AH']

scaler = MinMaxScaler()
df[num_cols] = scaler.fit_transform(df[num_cols])

df.head()
```

Out[21]:

	Date	Time	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC)	N
0	114	18	0.211864	0.511845	0.120981	0.185535	0.362097	0
1	114	19	0.161017	0.463029	0.088832	0.146226	0.312398	0
2	114	20	0.177966	0.541996	0.068528	0.139937	0.303659	0
3	114	21	0.177966	0.523331	0.061760	0.143082	0.308575	0
4	114	22	0.127119	0.448672	0.037225	0.100629	0.247406	0

In [22]:

```python
# Verify range [0, 1] after normalisation
df[num_cols].describe()
```

Out[22]:

	CO(GT)	PT08.S1(CO)	NMHC(GT)	C6H6(GT)	PT08.S2(NMHC)	93
count	9358.000000	9358.000000	9358.000000	9358.000000	9358.000000	
mean	0.173962	0.325078	0.179198	0.156967	0.303743	
std	0.111525	0.152750	0.054033	0.114815	0.142843	
min	0.000000	0.000000	0.000000	0.000000	0.000000	
25%	0.093220	0.211055	0.179198	0.070755	0.196614	
50%	0.173962	0.307251	0.179198	0.133648	0.295194	
75%	0.211864	0.412060	0.179198	0.212264	0.394320	
max	1.000000	1.000000	1.000000	1.000000	1.000000	

---

d. Error Correcting

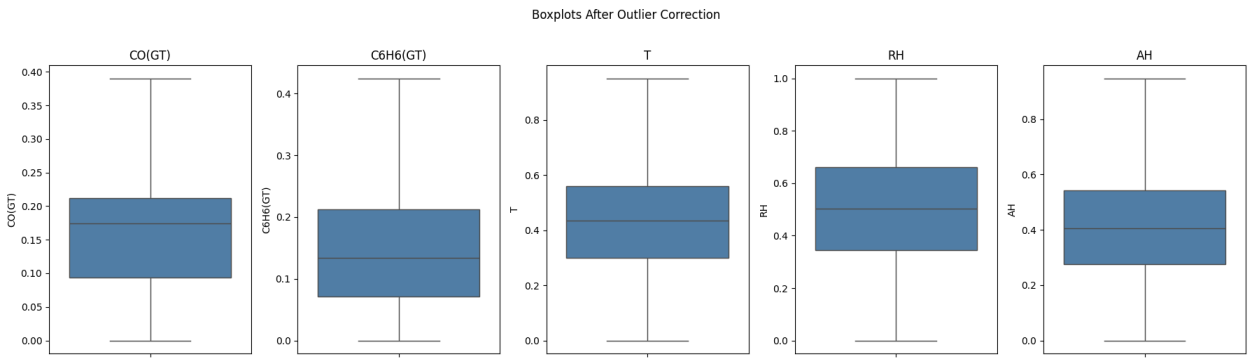
Outlier detection and removal using the IQR method.

In [23]:
```
def remove_outliers(col_series):
    Q1  = col_series.quantile(0.25)
    Q3  = col_series.quantile(0.75)
    IQR = Q3 - Q1
    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR
    return col_series.clip(lower=lower, upper=upper)  # clip instead of drop t
```

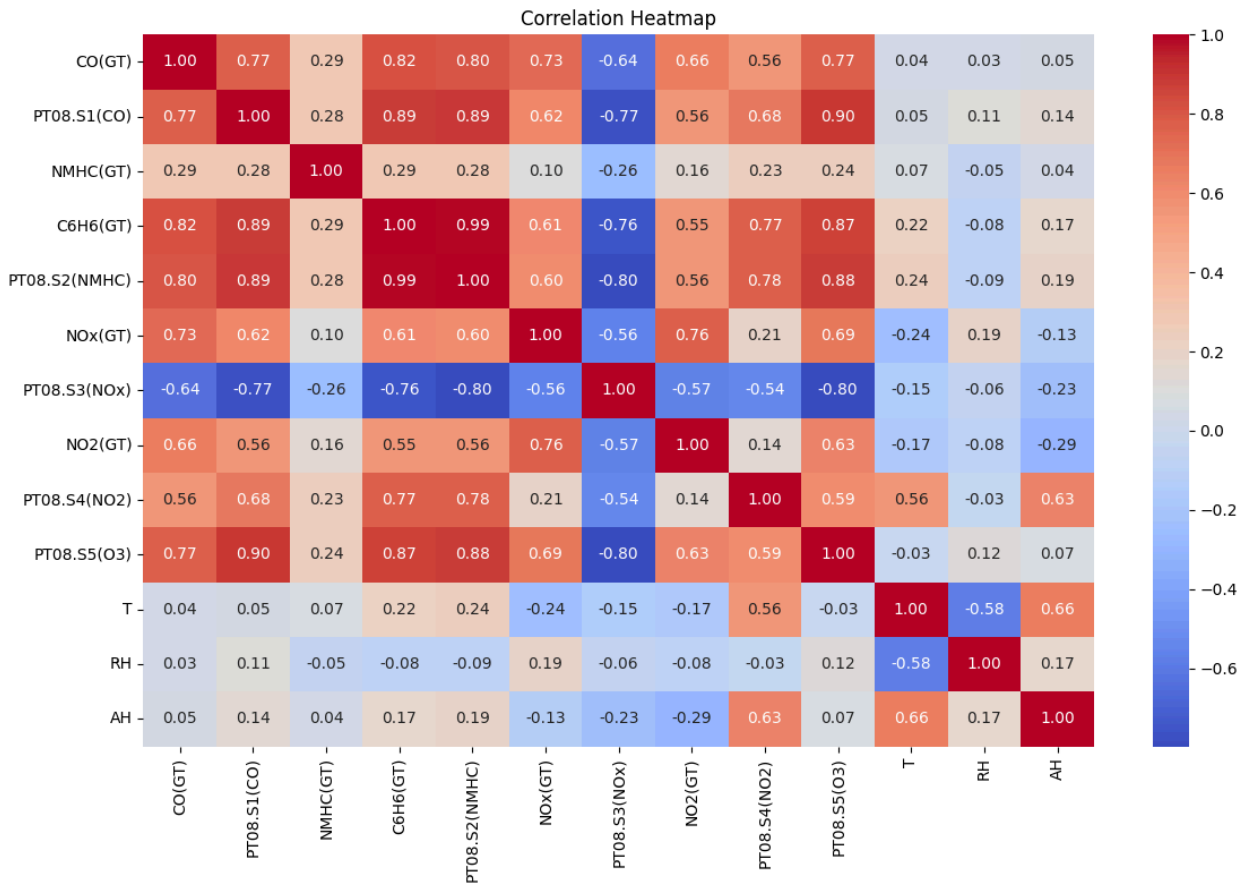
In [24]:
```
outlier_cols = ['CO(GT)', 'C6H6(GT)', 'T', 'RH', 'AH']

for col in outlier_cols:
    df[col] = remove_outliers(df[col])

print('Shape after outlier correction:', df.shape)
```

```
Shape after outlier correction: (9358, 15)
```

In [25]:
```
fig, axes = plt.subplots(1, len(outlier_cols), figsize=(18, 5))

for ax, col in zip(axes, outlier_cols):
    sns.boxplot(y=df[col], ax=ax, color='steelblue')
    ax.set_title(col)

plt.suptitle('Boxplots After Outlier Correction', y=1.02)
plt.tight_layout()
plt.show()
```



In [26]:
```
plt.figure(figsize=(12, 8))
sns.heatmap(df[num_cols].corr(), annot=True, fmt='.2f', cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.tight_layout()
plt.show()
```



---

e. Data Model Building

Predicting **CO(GT)** (Carbon Monoxide concentration) using sensor readings and environmental features with **Linear Regression**.

In [27]:
```
# Features and target
X = df[['PT08.S1(CO)', 'C6H6(GT)', 'T', 'RH', 'AH',
         'NOx(GT)', 'NO2(GT)', 'PT08.S5(O3)']]
y = df['CO(GT)']

print('Feature matrix shape:', X.shape)
print('Target vector shape :', y.shape)
```

```
Feature matrix shape: (9358, 8)
Target vector shape : (9358,)
```

In [28]:
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

print('Training set :', X_train.shape)
print('Test set     :', X_test.shape)
```

```
Training set : (7486, 8)
Test set     : (1872, 8)
```

In [29]:
```
# Train Linear Regression model
model = LinearRegression()
model.fit(X_train, y_train)
```

Out[29]:

▾ LinearRegression

Parameters

fit_intercept	True
copy_X	True
tol	1e-06
n_jobs	None
positive	False

In [30]:
```
# Model coefficients
coef_df = pd.DataFrame({'Feature': X.columns, 'Coefficient': model.coef_})
coef_df.sort_values('Coefficient', ascending=False)
```

Out[30]:

	Feature	Coefficient
1	C6H6(GT)	0.502290
5	NOx(GT)	0.207547
6	NO2(GT)	0.108976
0	PT08.S1(CO)	0.094824
3	RH	0.013199
4	AH	0.004392
2	T	0.003468
7	PT08.S5(O3)	-0.081487

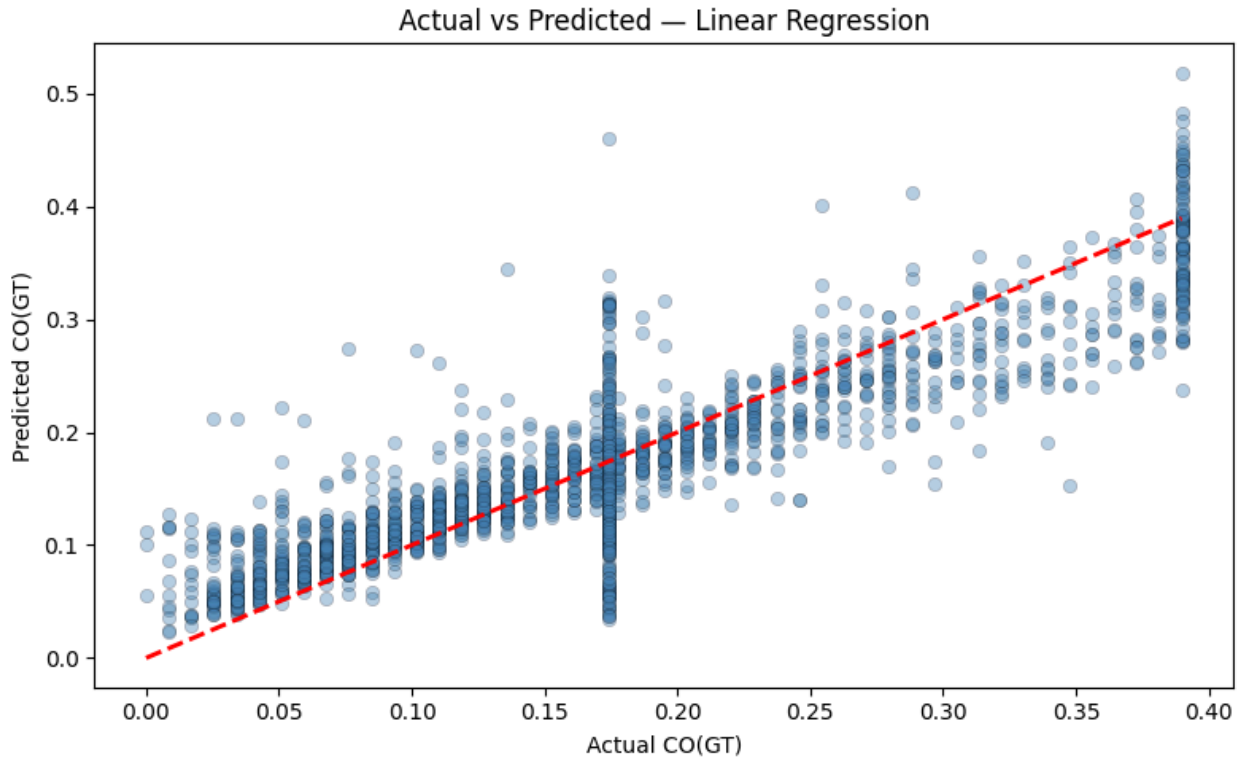
In [31]:

```
# Evaluate on test set
y_pred = model.predict(X_test)

mse  = mean_squared_error(y_test, y_pred)
rmse = np.sqrt(mse)
r2   = r2_score(y_test, y_pred)

print(f'MSE  : {mse:.4f}')
print(f'RMSE : {rmse:.4f}')
print(f'R²   : {r2:.4f}')
```

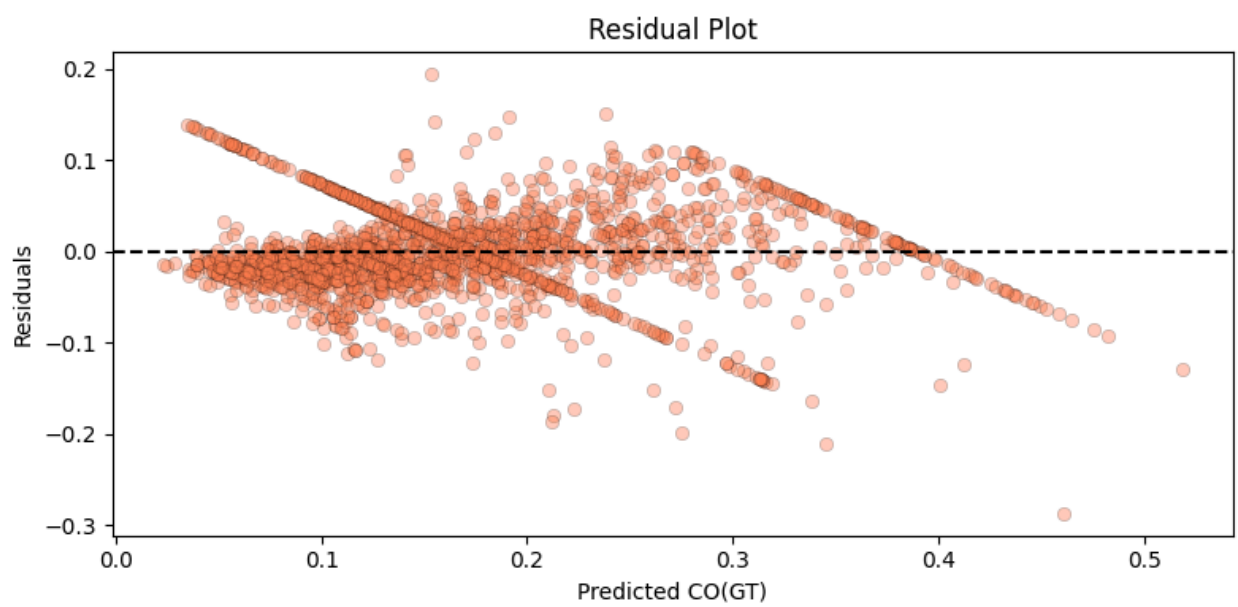
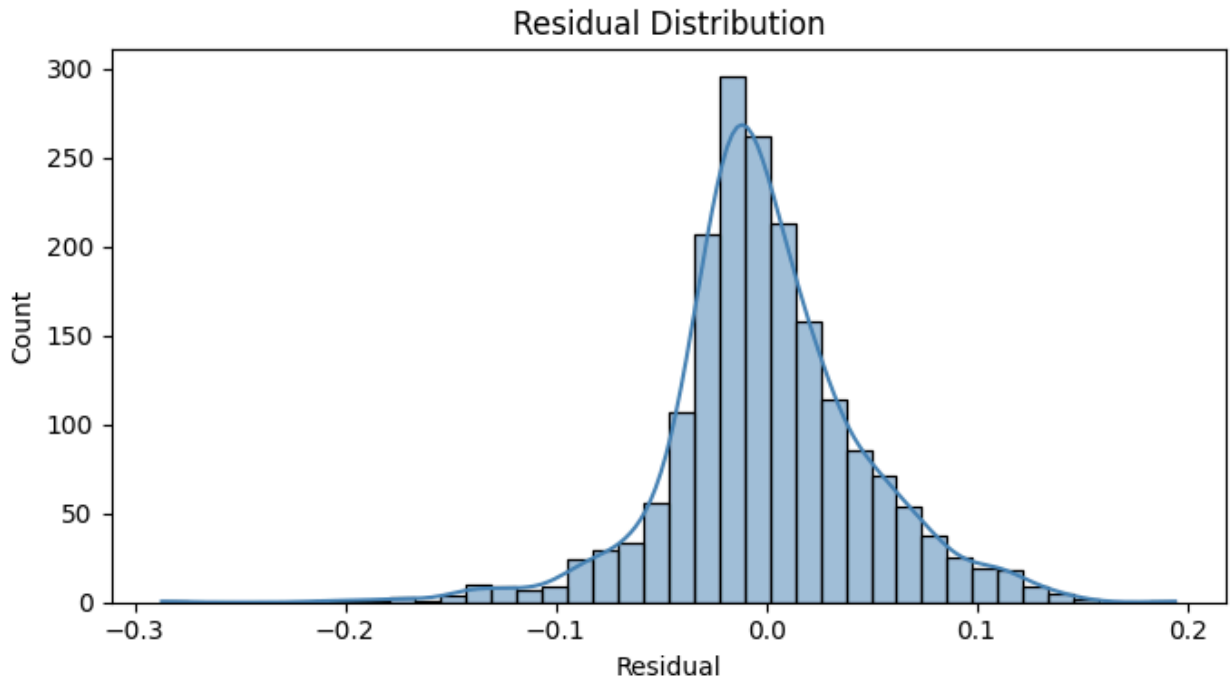
```
MSE  : 0.0022
RMSE : 0.0464
R²   : 0.7762
```

In [32]:

```
plt.figure(figsize=(8, 5))
plt.scatter(y_test, y_pred, alpha=0.4, color='steelblue', edgecolors='k', line
plt.plot([y_test.min(), y_test.max()], [y_test.min(), y_test.max()], 'r--', lw
plt.xlabel('Actual CO(GT)')
plt.ylabel('Predicted CO(GT)')
plt.title('Actual vs Predicted — Linear Regression')
plt.tight_layout()
plt.show()
```



In [33]:
```
residuals = y_test - y_pred

plt.figure(figsize=(8, 4))
plt.scatter(y_pred, residuals, alpha=0.4, color='coral', edgecolors='k', linew
plt.axhline(0, color='black', lw=1.5, linestyle='--')
plt.xlabel('Predicted CO(GT)')
plt.ylabel('Residuals')
plt.title('Residual Plot')
plt.tight_layout()
plt.show()
```



In [34]:
```
plt.figure(figsize=(7, 4))
sns.histplot(residuals, kde=True, color='steelblue', bins=40)
plt.xlabel('Residual')
plt.title('Residual Distribution')
plt.tight_layout()
plt.show()
```